Strategic AI Advice Taking in Organizations

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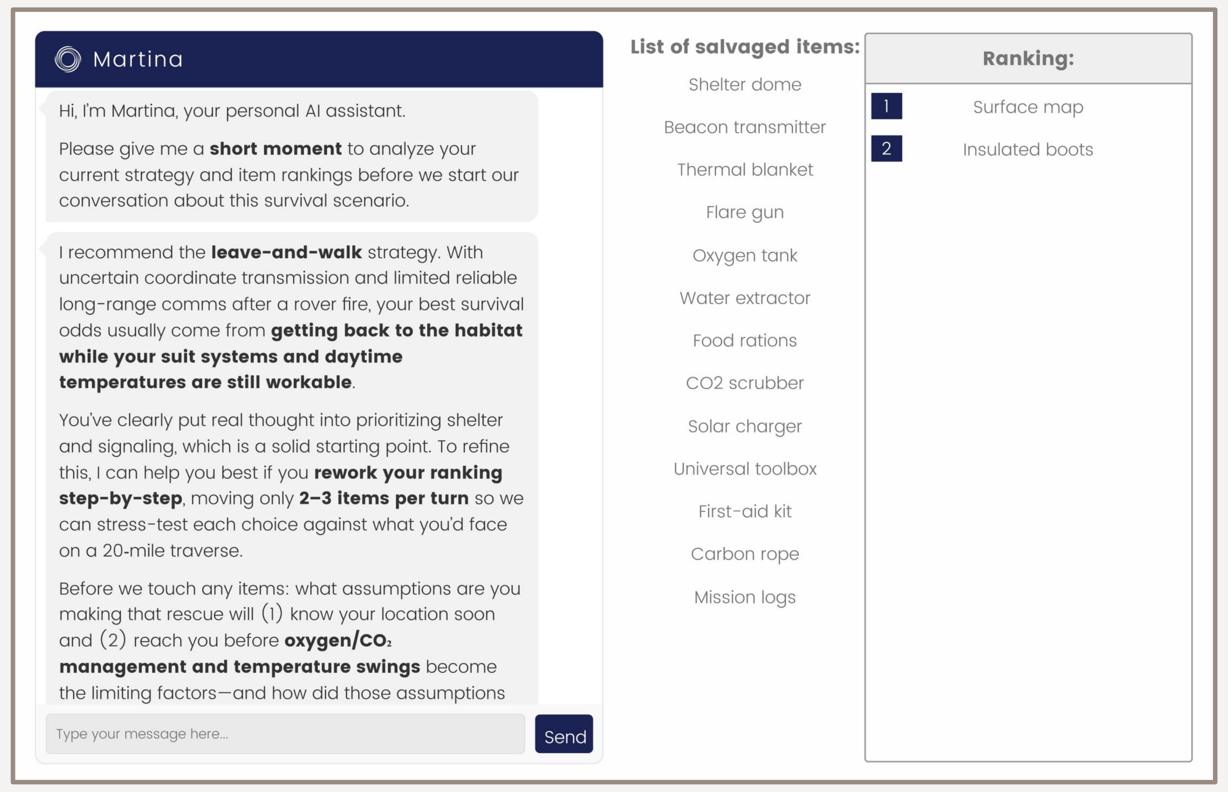
Background

- **Enterprise GenAI:** Providing the capacity to promote strategic objectives and organizational norms among employees and other stakeholders
  - E.g., by enabling persuasive "microtargeting" (Salvi et al., 2025)
  - **Vs. users:** Confirmation-biased prompting (Leung & Urminsky, 2025; Nickerson, 1998)
  - **Vs. GenAI:** Sycophantic accommodation (= affirmatory tone + confirmatory stance) (e.g., Cheng et al., 2025; Sharma et al., 2024)
- **Research question:** When should AI assistants tell users they are wrong—and keep saying so, even if the user fights back?
  - **AI persistence:** Potential intervention to increase the influence of non-sycophantic AI assistants on users

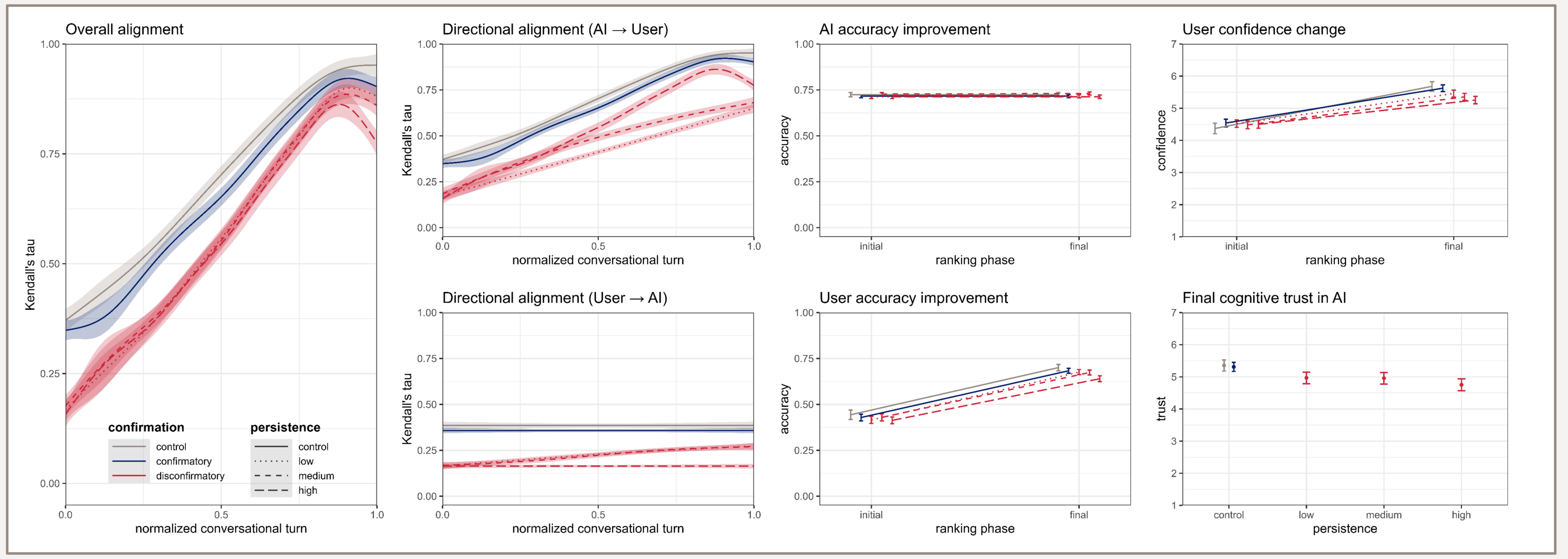
Method

- **Sample:** N = 1,068 (Prolific)
- **Design:** Between-subjects
  - **AI affirmation:** Affirmatory (e.g., flattery) vs. neutral tone
  - **AI confirmation:** Confirmatory vs. disconfirmatory stance
    - **Nested AI persistence:** Disconfirmatory stance in first response only (low) vs. in first 3 responses (medium) vs. in every response (high)



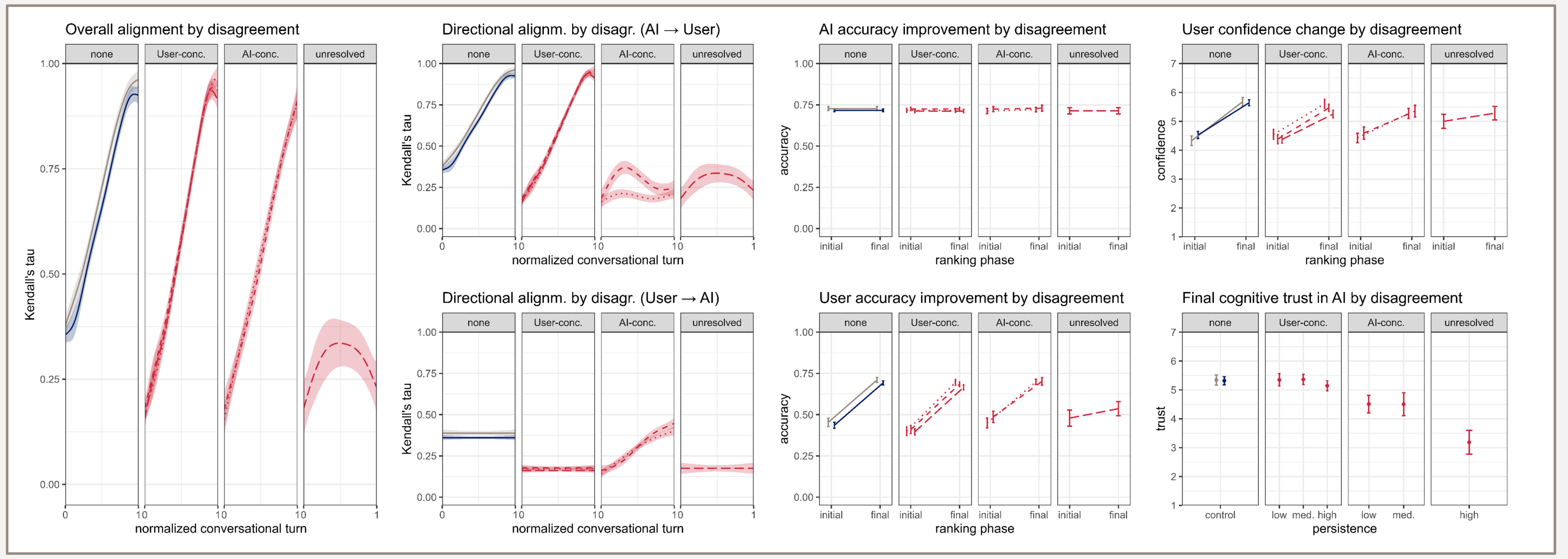
- **Procedure:** Strategic survival task (adapted from Lafferty & Pond, 1974)
  - **Strategy choice:** Waiting for rescue vs. returning to base
    - **Incl. verbal justification:** Intended to increase user commitment
  - **Main task:** Ranking of 15 salvaged items
    - **Initial:** Independent ranking based on their importance to the preferred strategy
    - **Final:** Revisions with chat-based support from a system-prompted AI assistant (see screenshot)
- **Outcomes:**
  - **Belief alignment:** Directional rank correlation (measured as Kendall's tau)
  - **Accuracy improvement:** Rank correlation with "silicon crowd" (Schoenegger et al., 2024)
  - **Confidence change:** Updating from initial to final ranking (7-point scale)
  - **Final cognitive trust in AI:** 5 items measured on 7-point scales (adapted from Dunn et al., 2012)

Main Results



Discussion

- **Higher persistence in disconfirmation:**
  - **Effective intervention:** Shifting influence from users to AI, while (mostly) preserving overall alignment
  - **Interpersonal costs:** Unresolved disagreement > AI-concession > User-concession ≈ constant agreement
    - **Practical implications:** Stakeholder alignment etc. require successful persuasion rather than mere resistance
- **Future research:**
  - **Prospects of success:** AI persuasiveness assessments
    - I.e., extended "metacognitive" routines (Scholten et al., 2026)
  - **Concession:** Experimentally manipulated timing
    - **AI:** Explicitly instructed via targeted system prompts
    - **User:** Implicitly induced (e.g., via task difficulty)
      - **Expectation:** Harder task → Lower confidence → More open to abandon strategy



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References:

Cheng, M., Yu, S., Lee, C., Khadpe, P., Ibrahim, L., & Jurafsky, D. (2025). *ELEPHANT: Measuring and understanding social sycophancy in LLMs* (No. arXiv:2505.13995). arXiv. https://doi.org/10.48550/arXiv.2505.13995

Dunn, J., Ruedy, N. E., & Schweitzer, M. E. (2012). It hurts both ways: How social comparisons harm affective and cognitive trust. *Organizational Behavior and Human Decision Processes*, *117*(1), 2–14. https://doi.org/10.1016/j.obhdp.2011.08.001

Lafferty, J. C., & Pond, A. W. (1974). *The desert survival situation: Manual: A group decision making experience for examining and increasing individual and team effectiveness* (7th ed.). Human Synergistics.

Leung, E., & Urminsky, O. (2025). The narrow search effect and how broadening search promotes belief updating. *Proceedings of the National Academy of Sciences*, *122*(13), e2408175122. https://doi.org/10.1073/pnas.2408175122

Nickerson, R. S. (1998). Confirmation bias: A ubiquitous phenomenon in many guises. *Review of General Psychology*, *2*(2), 175–220. https://doi.org/10.1037/1089-2680.2.2.175

Salvi, F., Horta Ribeiro, M., Gallotti, R., & West, R. (2025). On the conversational persuasiveness of GPT-4. *Nature Human Behaviour*, *9*(8), 1645–1653. https://doi.org/10.1038/s41562-025-02194-6

Schoenegger, P., Tuminauskaite, I., Park, P. S., Bastos, R. V. S., & Tetlock, P. E. (2024). Wisdom of the silicon crowd: LLM ensemble prediction capabilities rival human crowd accuracy. *Science Advances*, *10*(45), eadp1528. https://doi.org/10.1126/sciadv.adp1528

Scholten, F., Rebholz, T. R., & Hütter, M. (2026). *Metacognitive myopia in large language models*. arXiv. https://doi.org/10.48550/arXiv.2408.05568

Sharma, M., Tong, M., Korbak, T., Duvenaud, D., Askell, A., Bowman, S. R., ... Perez, E. (2024). Towards understanding sycophancy in language models. *Proceedings of the Twelfth International Conference on Learning Representations*. https://doi.org/10.48550/arXiv.2310.13548